Calculus Review

From Chapter 2 of the book “Dive into Deep Learning”

https://d2l.ai/

Outline

- Derivatives and differentiation
- Partial derivatives and gradients
- Chain Rule

Derivatives and Differentiation

- A *derivative* is the rate of change in a function with respect to changes in its arguments.
- Derivatives can tell us how rapidly a loss function would *increase* or *decrease* were we to increase or decrease each parameter by an infinitesimally small amount.
- Formally, for functions $f: \mathbb{R} \rightarrow \mathbb{R}$, that map from scalars to scalars, the derivative of f at a point x is defined as

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

- The term on the right hand side is called the limit, which tells us what the ratio between a perturbation h and the change in the function value $f(x+h) - f(x)$ converges to as we shrink its size to zero.

Derivatives and Differentiation

- When $f'(x)$ exists f is said to be differentiable at x; and when $f'(x)$ exists for all x on a set, e.g., the interval $[a, b]$, we say that f is differentiable on this set.
- Not all functions are differentiable, including many that we wish to optimize, including accuracy and the area under the receiving operating characteristic (AUC).
- However, because computing the derivative of the loss is a crucial step in nearly all algorithms for training deep neural networks, we often optimize a differentiable *surrogate* instead.
- We can interpret the derivative $f'(x)$ as the **instantaneous** rate of change of $f(x)$ with respect to x.

Derivatives and Differentiation

- Let's develop some intuition with an example. Define

$$u = f(x) = 3x^2 - 4x$$

Setting $x = 1$, $\frac{f(x+h)-f(x)}{h}$ approaches 2 as h approaches 0. While this experiment lacks the rigor of a mathematical proof, we will soon see that indeed $f'(1) = 2$.

There are several equivalent notational conventions for derivatives. Given $y = f(x)$, the following expressions are equivalent:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x), \tag{2.4.2}$$

where the symbols $\frac{d}{dx}$ and D are *differentiation operators*. Below, we present the derivatives of some common functions:

Derivatives and Differentiation

$$\frac{d}{dx}C = 0 \qquad \text{for any constant } C$$

$$\frac{d}{dx}x^n = nx^{n-1} \qquad \text{for } n \neq 0$$

$$\frac{d}{dx}e^x = e^x$$

$$\frac{d}{dx}\ln x = x^{-1}$$

Derivatives and Differentiation

Functions composed from differentiable functions are often themselves differentiable. The following rules come in handy for working with compositions of any differentiable functions f and g, and constant C.

$$\begin{aligned}
\frac{d}{dx}[Cf(x)] &= C\frac{d}{dx}f(x) && \text{Constant multiple rule} \\
\frac{d}{dx}[f(x)+g(x)] &= \frac{d}{dx}f(x)+\frac{d}{dx}g(x) && \text{Sum rule} \\
\frac{d}{dx}[f(x)g(x)] &= f(x)\frac{d}{dx}g(x)+g(x)\frac{d}{dx}f(x) && \text{Product rule} \\
\frac{d}{dx}\frac{f(x)}{g(x)} &= \frac{g(x)\frac{d}{dx}f(x)-f(x)\frac{d}{dx}g(x)}{g^2(x)} && \text{Quotient rule}
\end{aligned} \tag{2.4.4}$$

Derivatives and Differentiation

Using this, we can apply the rules to find the derivative of $3x^2 - 4x$ via

$$\frac{d}{dx}[3x^2 - 4x] = 3\frac{d}{dx}x^2 - 4\frac{d}{dx}x = 6x - 4. \tag{2.4.5}$$

Plugging in $x = 1$ shows that, indeed, the derivative is 2 at this location. Note that derivatives tell us the *slope* of a function at a particular location.

Partial Derivatives and Gradients

- Thus far, we have been differentiating functions of just one variable. In deep learning, we also need to work with functions of *many* variables.
- We briefly introduce notions of the derivative that apply to such *multivariate* functions.

Let $y = f(x_1, x_2, \ldots, x_n)$ be a function with n variables. The *partial derivative* of y with respect to its ith parameter x_i is

$$\frac{\partial y}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h}. \tag{2.4.6}$$

To calculate $\frac{\partial y}{\partial x_i}$, we can treat $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n$ as constants and calculate the derivative of y with respect to x_i. The following notation conventions for partial derivatives are all common and all mean the same thing:

$$\frac{\partial y}{\partial x_i} = \frac{\partial f}{\partial x_i} = \partial_{x_i} f = \partial_i f = f_{x_i} = f_i = D_i f = D_{x_i} f. \tag{2.4.7}$$

Partial Derivatives and Gradients

We can concatenate partial derivatives of a multivariate function with respect to all its variables to obtain a vector that is called the *gradient* of the function. Suppose that the input of function $f : \mathbb{R}^n \to \mathbb{R}$ is an n-dimensional vector $\boldsymbol{x} = [x_1, x_2, \ldots, x_n]^\top$ and the output is a scalar. The gradient of the function f with respect to $\boldsymbol{x}$ is a vector of n partial derivatives:

$$\nabla_{\boldsymbol{x}} f(\boldsymbol{x}) = [\partial_{x_1} f(\boldsymbol{x}), \partial_{x_2} f(\boldsymbol{x}), \ldots \partial_{x_n} f(\boldsymbol{x})]^\top . \tag{2.4.8}$$

When there is no ambiguity, $\nabla_{\boldsymbol{x}} f(\boldsymbol{x})$ is typically replaced by $\nabla f(\boldsymbol{x})$. The following rules come in handy for differentiating multivariate functions:

- For all $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ we have $\nabla_{\boldsymbol{x}} \boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}^\top$ and $\nabla_{\boldsymbol{x}} \boldsymbol{x}^\top \boldsymbol{A} = \boldsymbol{A}$.
- For square matrices $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ we have that $\nabla_{\boldsymbol{x}} \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} = (\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{x}$ and in particular $\nabla_{\boldsymbol{x}} \|\boldsymbol{x}\|^2 = \nabla_{\boldsymbol{x}} \boldsymbol{x}^\top \boldsymbol{x} = 2\boldsymbol{x}$.

Similarly, for any matrix $\boldsymbol{X}$, we have $\nabla_{\boldsymbol{X}} \|\boldsymbol{X}\|_F^2 = 2\boldsymbol{X}$.

Chain Rule

- In deep learning, the gradients of concern are often difficult to calculate because we are working with deeply nested functions (of functions (of functions…)). Fortunately, the chain rule takes care of this.

Returning to functions of a single variable, suppose that $y = f(g(x))$

and that the underlying functions $y = f(u)$ and $u = g(x)$ are both differentiable. The chain rule states that

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}. \tag{2.4.9}$$

The Chain Rule

Turning back to multivariate functions, suppose that $y = f(\boldsymbol{u})$ has variables $u_1, u_2, \ldots, u_m$, where each $u_i = g_i(\boldsymbol{x})$ has variables $x_1, x_2, \ldots, x_n$, i.e., $\boldsymbol{u} = g(\boldsymbol{x})$. Then the chain rule states that

$$\frac{\partial y}{\partial x_i} = \frac{\partial y}{\partial u_1}\frac{\partial u_1}{\partial x_i} + \frac{\partial y}{\partial u_2}\frac{\partial u_2}{\partial x_i} + \ldots + \frac{\partial y}{\partial u_m}\frac{\partial u_m}{\partial x_i} \text{ and thus } \nabla_{\boldsymbol{x}} y = \boldsymbol{A}\nabla_{\boldsymbol{u}} y, \qquad (2.4.10)$$

where $\boldsymbol{A} \in \mathbb{R}^{n \times m}$ is a *matrix* that contains the derivative of vector $\boldsymbol{u}$ with respect to vector $\boldsymbol{x}$. Thus, evaluating the gradient requires computing a vector-matrix product. This is one of the key reasons why linear algebra is such an integral building block in building deep learning systems.

Exercises – Homework

What does it mean that $f'(x) = 0$ for some x? Give an example of a function f and a location x for which this might hold.

Find the gradient of the function $f(\boldsymbol{x}) = 3x_1^2 + 5e^{x_2}$.

Can you write out the chain rule for the case where $u = f(x, y, z)$ and $x = x(a, b)$, $y = y(a, b)$, and $z = z(a, b)$?